

Notes: Sun & Tong (JFE 2003)

China Share Issue Privatization: The Extent of Its Success

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1 Big picture and research question

1.1 Core question

China’s reform during the 1990s often took the form of *partial* privatization: SOEs issued a minority of shares to the public (listed equity) while the state typically retained a large stake. The paper asks:

Does share issue privatization (listing and selling equity to outside investors) improve firm performance in China, and through which ownership/governance channels?

1.2 What the paper contributes

- It brings large-sample evidence on Chinese SIP (634 firms listed 1994–1998) using a *before–after* design around the listing year (“year 0”).
- It uses outcome measures that are more robust to China’s accounting reforms and IPO-related balance-sheet jumps (e.g., *real levels* of profits and output, not ROA/ROE).
- It documents that SIP is associated with:
 - higher *real profits* and much higher *real sales/output*,
 - but *lower profitability ratios* and *weaker debt-service/coverage* measures,
 - and modestly higher labor productivity (sales per employee).
- It links heterogeneity to ownership structure: state ownership is (weakly) value-reducing, while legal-person ownership is value-enhancing.

1.3 One sentence

SIP in China appears to raise scale/output and absolute earnings, but does not cleanly deliver profitability improvements; governance still matters because the state often keeps large (non-tradable) control blocks.

2 Institutional setting: what “privatization” means in China

2.1 Share classes and who owns what

A Chinese listed firm’s equity can be separated into (i) *tradable shares* and (ii) *non-tradable shares*:

- **State shares (ST):** owned by central/local governments or state agencies; typically *non-tradable* in this period.
- **Legal person shares (LP):** owned by domestic institutions (often corporatized SOEs, state-controlled holding companies, or other legal entities); also largely *non-tradable*.
- **Tradable shares:** held by public investors.

- **A-shares:** domestic RMB-denominated.
- **B/H-shares:** foreign-accessible shares (foreign currency or Hong Kong listing); summarized by the paper as **FRN**.

2.2 Two degrees of “privatization” in the paper

The paper distinguishes the size of the post-listing government stake:

- **Revenue privatization:** government retains *more than 50%* of shares (keeps voting control).
- **Control privatization:** government retains *less than 50%* of shares (control is more contestable).

2.3 Why ownership composition should matter

The intuition is a standard agency/governance one:

- The state may pursue political objectives (employment, soft budgets, local protection) and may weaken incentives for profit maximization.
- “Legal persons” may behave more like concentrated blockholders with monitoring incentives, *especially* if they are more commercialized and face harder budget constraints than government bureaus.
- Foreign investors are often minority holders and face information/monitoring frictions; their influence may be limited unless their stake is economically meaningful.

3 Data and construction of outcome variables

3.1 Sample

- 634 firms listed on Shanghai and Shenzhen stock exchanges during 1994–1998.
- Pre-IPO data for up to 3 years come from IPO prospectuses; post-IPO data come from databases and annual reports.
- Financial firms are excluded; pre-1994 listings are excluded due to major accounting changes in China in 1993–1994.
- The “event time” convention: listing year is $t = 0$, and the analysis focuses on $t \in \{-3, -2, -1, +1, +2, +3\}$.

3.2 Why the paper avoids ROA/ROE

IPO proceeds and restructuring can mechanically change assets and equity, so ratios like ROA/ROE can fall even if the firm becomes more efficient. Hence the paper emphasizes *levels* (real profits and real sales) and “less mechanical” ratios such as return on sales.

3.3 Real and normalized performance series

For level variables (profits and sales), the paper (i) deflates into real terms and (ii) normalizes to 1 in year 0.

Let $X_{i,t}$ denote a nominal level variable (net profit, EBIT, or sales). Let P_t be a price index. Define real values $X_{i,t}^{real} = X_{i,t}/P_t$ and the *normalized* series:

$$\tilde{X}_{i,t} \equiv \frac{X_{i,t}^{real}}{X_{i,0}^{real}}. \quad (1)$$

Then define three-year averages:

$$\bar{\tilde{X}}_{i,-} \equiv \frac{1}{3} \sum_{\tau=-3}^{-1} \tilde{X}_{i,\tau}, \quad \bar{\tilde{X}}_{i,+} \equiv \frac{1}{3} \sum_{\tau=+1}^{+3} \tilde{X}_{i,\tau}, \quad (2)$$

and the change:

$$\Delta \tilde{X}_i \equiv \bar{\tilde{X}}_{i,+} - \bar{\tilde{X}}_{i,-}. \quad (3)$$

3.4 Statistical tests used in Tables 1–5

- **Within-firm change (before vs after):** Wilcoxon signed-rank test on Δ .
- **Between-group differences:** Wilcoxon rank-sum test comparing Δ across two groups.
- **Share of firms improving:** a binomial/proportion Z test based on the fraction with $\Delta > 0$.

4 Before–after evidence on performance, output, leverage, and labor

4.1 Table 1: absolute earnings increase (real net profit and real EBIT)

Variables.

- **RNP:** real net profit (normalized to 1 in year 0).
- **REBIT:** real EBIT (normalized to 1 in year 0).

Main findings (whole sample).

- Median RNP rises from 0.7277 (pre) to 1.0925 (post), median change +0.3648 with a strong signed-rank statistic (Wilcoxon $Z \approx 9.80$).
- Median REBIT rises from 0.8101 to 1.1338, median change +0.3237 (Wilcoxon $Z \approx 9.11$).
- A large majority of firms show positive changes (RNP: 433 up vs 200 down; REBIT: 413 up vs 193 down).

Cross-sectional patterns (selected).

- *Regulated industries* improve more: median Δ RNP about 0.588 vs 0.344 in nonregulated; the between-group difference is significant.
- *Control privatization* (state < 50%) shows larger improvements than revenue privatization (state > 50%), consistent with the view that selling voting control matters.
- *Later cohorts* (1996–1998 listings) show stronger earnings improvements than early cohorts (1994–1995), especially for net profit.

Table 1

Comparison of absolute earnings before and after privatization

The table presents, in various samples, the number of observations, the mean and median values of real net profit (RNP) and real EBIT (REBIT) for the average of three years pre- and post-privatisation periods, the mean and median change in RNP and REBIT value (post-pre), and the tests of significance of the median change. The Wilcoxon Z-test is employed to test for any significant change in the median value (paired observation), and for any significant difference in changes between any two groups. The last column shows the number of positive versus negative changes and the proportion Z-test on if the proportion of positive change is greater than 50%.

Variable	Sample	Obs.	Median (mean) before	Median (mean) after	Median (mean) change	Wilcoxon test		+ ve/–ve Ratio
						Paired obs.	Bet. gps.	
RNP	Whole sample (–3 to +3)	634	0.7277 (0.8967)	1.0925 (1.0379)	0.3648 (0.1412)	9.8020**		433/201 (13.5128)**
	SHSE vs SZSE	314	0.7060 (0.8559)	1.1046 (0.8954)	0.3986 (0.0395)	7.8635**	1.3581	221/93 (11.1175)**
	Reg. ind. vs non-reg. ind.	58 576	0.7246 (0.7125) 0.7292 (0.9153)	1.1972 (1.3591) 1.0797 (1.0056)	0.4726 (0.6466) 0.3505 (0.0903)	4.8894** 8.7260**	2.1379**	46/12 (8.6652)** 387/189 (11.7951)**
	Rev. privatization vs control privatization	158 476	0.7659 (0.9849) 0.7198 (0.8784)	1.0009 (0.9700) 1.1284 (1.0604)	0.2350 (0.0181) 0.4086 (0.1821)	3.2355** 9.4227**	2.3156**	99/59 (4.1091) 334/142 (13.4825)
	Foreign share vs no foreign share	40 594	0.7569 (0.9849) 0.7271 (0.8908)	0.8989 (0.8799) 1.1132 (1.0486)	0.1420 (–0.1050) 0.3861 (0.1578)	0.1203 10.2105**	2.7544**	22/18 (0.6904) 411/183 (14.0078)**
	Firms listed in 94–95 vs firms listed in 96–98	124 510	0.7229 (0.9404) 0.7279 (0.8861)	0.8173 (0.5890) 1.1522 (1.1471)	0.0944 (–0.3514) 0.4243 (0.2610)	8.7260** 11.1211**	4.5642**	67/57 (0.9699) 366/144 (15.6567)**
	Whole sample (–3 to +3)	606	0.8149 (0.9000)	1.1386 (1.2722)	0.3237 (0.3723)	9.1090**		413/193 (13.0624)**
	SHSE vs SZSE	293 313	0.8092 (0.8410) 0.8180 (0.9552)	1.1554 (1.2372) 1.0922 (1.3050)	0.3462 (0.3963) 0.2742 (0.3498)	7.3783** 5.5271**	1.6636*	204/89 (10.1541)** 209/104 (8.3999)**
	Reg. ind. vs	55	0.8182 (0.7820)	1.2489 (1.4635)	0.4307 (0.6815)	4.9021**	2.1791**	41/14 (6.1734)**

Variable	Sample	Obs.	Median (mean) before	Median (mean) after	Median (mean) change	Wilcoxon test		+ ve/–ve Ratio
						Paired obs.	Bet. gps.	
	non-reg.ind.	551	0.8145 (0.9117)	1.1161 (1.2532)	0.2616 (0.3414)	8.0012**		372/179 (11.8422)**
	Rev. privatization vs control privatization	148 458	0.8825 (1.0627) 0.8000 (0.8474)	1.0114 (1.3279) 1.1540 (1.2543)	0.1289 (0.2652) 0.3540 (0.4069)	2.2389** 9.1649**	2.2847**	91/57 (3.5193) 342/136 (13.3587)**
	Foreign share vs no foreign share	36 570	0.8618 (0.9400) 0.8114 (0.8974)	0.8653 (1.0089) 1.1476 (1.2889)	0.0035 (0.0689) 0.3362 (0.3914)	0.0056 0.3590**	2.0234**	18/18 (0) 395/175 (13.8319)**
	Firms listed in 94–95 vs firms listed in 96–98	100 506	0.8798 (0.9800) 0.7988 (0.8841)	0.7507 (0.9147) 1.1786 (1.3429)	–0.1291 (–0.0653) 0.3798 (0.4588)	1.5454 10.8329**	21.1262**	49/51 (–0.1951) 364/142 (15.7854)**

* (**) Denotes significance at the 10% (5%) level (two-tails).

4.2 Table 2: profitability ratios decline on average (ROS and EBIT/Sales) Variables.

- $ROS \equiv \frac{\text{net income}}{\text{sales}}$.
- $EBITS \equiv \frac{\text{EBIT}}{\text{sales}}$.

Main findings (whole sample).

- Median ROS falls from 0.1234 to 0.1142, median change -0.0092 (Wilcoxon $Z \approx 3.05$).
- Median EBITs falls from 0.1929 to 0.1751, median change -0.0178 (Wilcoxon $Z \approx 3.97$).

- The number of firms with declining ratios exceeds the number improving (e.g., ROS: 279 up vs 355 down).

Table 2

Comparison of the percentage profitability of privatised firms

The table presents, in various samples, the number of observations, the mean and median values of return on sales (ROS) and return on RBIT (EBIT/sales) for the average of three years pre- and post-privatisation periods, the mean and median change in ROS and EBIT/sales value (post-pre), and the tests of significance of the median change. The Wilcoxon Z-test is employed to test for any significant change in the median value (paired observation), and for any significant difference in changes between any two groups. The last column shows the number of positive versus negative changes and the proportion Z-test on if the proportion of positive change is greater than 50%.

Sample	Obs.	Variable	Subsample	Median (mean) before	Median (mean) after	Median (mean) change	Wilcoxon test		+ve/-ve Ratio
							Paired obs.	Bet. groups	
Whole sample (-3 to +3)	634	ROS		0.1234 (0.1647)	0.1142 (0.1136)	-0.0092 (-0.0511)	3.0547**		279/355 (-2.6737)**
	614	EBIT/ sales		0.1728 (0.2119)	0.1550 (0.1636)	-0.0178 (-0.0482)	3.9729**		245/369 (-4.0743)**
SHSE vs SZSE	314	ROS	SH	0.1139 (0.1650)	0.1149 (0.1075)	0.001 (-0.0575)	1.6366	0.1360	137/177 (-1.9828)**
	320		SZ	0.1293 (0.1645)	0.1132 (0.1196)	-0.0161 (-0.0449)	2.6649**		142/178 (-1.7947)*
	299	EBIT/ sales	SH	0.1641 (0.2080)	0.1543 (0.1575)	-0.0098 (-0.0505)	2.4668**	0.7455	128/171 (-2.1479)**
	315		SZ	0.1851 (0.2155)	0.1560 (0.1694)	-0.0291 (-0.0461)	3.1398**		117/198 (-3.5027)**
Regulated industry vs	58	ROS	Regulated	0.1677 (0.2059)	0.1919 (0.1700)	0.0242 (-0.0359)	0.2954	0.9285	30/28 (0.2695)
	576		Non- regulated	0.1180 (0.1606)	0.1097 (0.1079)	-0.0083 (-0.0527)	3.0739**		249/327 (-2.8336)**
non-regulated industry	55	EBIT/ sales	Regulated	0.2265 (0.2579)	0.2558 (0.2357)	0.0293 (-0.0221)	0.2630	1.2643	25/30 (-0.6099)
	559		Non- regulated	0.1680 (0.2073)	0.1515 (0.1565)	-0.0165 (-0.0508)	4.0050**		220/339 (-4.051)**
Revenue privatisation vs	158	ROS	Revenue	0.1124 (0.1531)	0.1052 (0.1066)	-0.0072 (-0.0465)	2.0940**	1.0304	60/98 (-2.358)**
	476		Control	0.1240 (0.1686)	0.1182 (0.1159)	-0.0058 (-0.0527)	2.3033**		219/257 (-1.6061)
control privatisation	152	EBIT/ sales	Revenue	0.1700 (0.2087)	0.1495 (0.1529)	-0.0205 (-0.0558)	2.7711**	0.1703*	54/98 (-2.6405)**
	462		Control	0.1736 (0.2129)	0.1582 (0.1671)	-0.0154 (-0.0457)	2.9736**		191/271 (-3.1213)**
With foreign share	40	ROS	Foreign	0.1087 (0.1558)	0.0846 (0.0571)	-0.0241 (-0.0987)	1.9197*	1.9910*	9/31 (-1.8507)*

Table 2 (continued)

Sample	Obs.	Variable	Subsample	Median (mean) before	Median (mean) after	Median (mean) change	Wilcoxon test		+ve/-ve Ratio
							Paired obs.	Bet. groups	
vs	594		Local	0.1245 (0.1653)	0.1171 (0.1174)	-0.0074 (-0.0479)	2.6765***		270/324 (-2.0209)**
	36	EBIT/ sales	Foreign	0.1533 (0.1950)	0.1061 (0.0985)	-0.0472 (-0.0965)	1.8526*	1.2109	11/25 (-1.5261)
Without foreign share	578		Local	0.1756 (0.2129)	0.1558 (0.1677)	-0.0198 (-0.0452)	3.6296**		234/344 (-3.7703)**
Firms listed in 94-95 vs	124	ROS	94, 95	0.0994 (0.1309)	0.0681 (0.0771)	-0.0313 (-0.0537)	2.7772**	1.6965*	47/77 (-2.0963)**
	510		96, 97, 98	0.1290 (0.1730)	0.1214 (0.1225)	-0.0076 (-0.0505)	2.0232**		232/278 (-1.859)*
Firms listed in 96-98	107	EBIT/ sales	94, 95	0.1475 (0.1796)	0.1026 (0.1184)	-0.0449 (-0.0612)	2.8218**	1.8394*	35/72 (-2.4822)**
	507		96, 97, 98	0.1834 (0.2187)	0.1644 (0.1732)	-0.0190 (-0.0455)	3.0666**		210/297 (-3.2458)**

() Denotes significance at the 10% (5%) level (two-tails).

Interpretation. The profitability ratios decline even though levels of profits increase, because *sales/output expands even more strongly* (Table 3), and because post-IPO restructuring and competition may compress margins.

Heterogeneity that matters.

- The decline is driven by *nonregulated industries*; regulated industries show much weaker declines and even positive median changes.
- Early cohorts (1994–1995) exhibit larger profitability declines than later cohorts (1996–1998), consistent with changing policy regimes and/or learning-by-doing in the reform process.

4.3 Table 3: real sales/output increases strongly

Variable. Real sales (output) normalized to 1 in year 0.

Main findings (whole sample).

- Median real sales increases from 0.8827 (pre) to 1.2489 (post), median change +0.3662 with a very large Wilcoxon statistic ($Z \approx 15.72$).

- The majority of firms expand output (about 475 up vs 159 down).

Heterogeneity.

- *Control privatization* shows significantly larger sales growth than revenue privatization (median Δ about 0.419 vs 0.230).
- Firms with foreign shares have smaller sales growth than those without, and early cohorts show almost no median sales growth.

Table 3

Results for the output (real sales) of privatized firms

The table presents, in various samples, the number of observations, the mean and median values of real sales for the average of three years pre- and post-privatization periods, the mean and median change in real sales value (post-pre), and the tests of significance of the median change. The real sales is standardized to one in year 0 (the year of privatization). The Wilcoxon Z-test is employed to test for any significant change in the median value (paired observation), and for any significant difference in changes between any two groups. The last column shows the number of positive versus negative changes and the proportion Z-test on if the proportion of positive change is greater than 50%.

Sample	Obs.	Variable	Subsample	Median (mean) before	Median (mean) after	Median (mean) change	Paired obs.	Wilcoxon test	+ve/-ve Ratio (Prop. Z)
Whole sample	633	Real sales		0.8827 (1.2489)	0.5662 (0.9648)	15.7169**	475/158		
SHSE vs SZSE	314	Real sales	SH	0.8829 (1.2413)	0.5584 (0.9048)	11.4006**	0.8457	236/78	
Regulated industry vs Non-regulated industry	58	Real sales	Regulated	0.8870 (1.4218)	0.5348 (0.9299)	5.7322**	1.5741	45/13	
Revenue privatization vs Control privatization	475	Real sales	Non-regulated	0.8827 (1.2260)	0.3433 (0.9155)	14.6547**	430/145		
With foreign share vs Without foreign share	40	Real sales	Control	0.8732 (1.2026)	0.4108 (0.9355)	14.7816**	368/107		
Firms listed in 94-95 vs Firms listed in 96-98	593	Real sales	Local	0.9077 (1.1208)	0.2131 (0.9355)	2.7761**	1.7709*	25/15	
Firms listed in 124 94-95 vs Firms listed in 509 96-98	124	Real sales	Early privatization	0.9403 (1.3085)	0.9743 (1.4600)	0.0338 (0.9805)	1.6069*	68/56	
	509	Real sales	Late privatization	0.8738 (1.2062)	0.4332 (0.9393)	16.6443**	407/102		

(*) Denotes significance at the 10% (5%) level (two-tail).

Table 4

Results for the leverage of privatized firms

The table presents, in various samples, the number of observations, the mean and median values of the time-interest-earned (TIE) and cash flow to total debt ratio (OCF/TD) for the average of three years pre- and post-privatization periods, the mean and median change in the output value (post-pre), and the tests of significance of the median change. The Wilcoxon Z-test is employed to test for any significant change in the median value (paired observation), and for any significant difference in changes between any two groups. The last column shows the number of positive versus negative changes and the proportion Z-test on if the proportion of positive change is greater than 50%.

Sample	Variable	N	Median (mean) before	Median (mean) after	Median (mean) change	Paired obs.	Wilcoxon test	+ve/-ve Ratio (Prop. Z)
Whole sample	OCF/TD	288	0.2360 (0.3521)	0.1800 (0.2328)	-0.0560 (-0.1193)	5.3551**	94/194	
	TIE	612	6.2485 (-7.4965)	4.7340 (6.0356)	-1.5145 (-13.5321)	4.022**	278/334	
Firms listed on SHSE vs Firms listed on SZSE	OCF/TD	167	0.2369 (0.3821)	0.1787 (0.2394)	-0.0582 (-0.1428)	3.9788**	57/110	
Reg. ind. vs Non-reg. ind.	OCF/TD	32	0.3065 (0.5130)	0.2091 (0.2678)	-0.0974 (-0.2452)	1.9039*	8/24	
Firms not in reg. ind. vs Firms listed in reg. ind.	OCF/TD	558	6.8290 (-32.7057)	5.5150 (-17.6758)	-1.3140 (-18.0479)	0.765	0.180	138/178
Revenue privatization vs Control privatization	OCF/TD	220	0.2286 (0.4016)	0.2032 (0.2395)	-0.0254 (-0.1641)	2.5091**	0.2041	27/41
Firms with foreign share vs Firms without foreign share	OCF/TD	13	0.2378 (0.2697)	0.1533 (0.1500)	-0.0845 (-0.1190)	2.4102**	1.4041	1/12

(*) Denotes significance at the 10% (5%) level (two-tail).

Table 4 (continued)

Sample	Variable	N	Median (mean) before	Median (mean) after	Median (mean) change	Paired obs.	Wilcoxon test	+ve/-ve Ratio (Prop. Z)
Firms without foreign share	TIE	36	7.2707 (14.3914)	2.5151 (-1.1863)	-4.7556 (-15.5776)	2.157**	16/20	
Firms listed in 94-95 vs Firms listed in 96-98	OCF/TD	48	0.2142 (0.2414)	0.1444 (0.1899)	-0.0698 (-0.0515)	4.5993**	17/31	
	TIE	107	7.5234 (0.5023)	3.7915 (-1.3843)	-3.7320 (-1.8879)	2.979**	40/67	

(*) Denotes significance at the 10% (5%) level (two-tail).

4.4 Table 4: debt-service capacity weakens (higher effective leverage)

Variables.

- $OCF/TD \equiv \frac{\text{operating cash flow}}{\text{total debt}}$.
- $TIE \equiv \frac{EBIT}{\text{interest expense}}$ (times-interest-earned).

Main findings.

- OCF/TD declines: median falls from 0.2360 to 0.1800; median change -0.0560 (Wilcoxon $Z \approx 5.36$).
- TIE declines: median falls from 6.2485 to 4.7340; median change -1.5145 (Wilcoxon $Z \approx 4.02$).

Interpretation. Even if output expands, firms appear to rely more on debt financing and/or face higher interest burdens. Thus, the “privatization” episode is *not* purely de-leveraging; it may coincide with balance-sheet expansion and additional borrowing.

4.5 Table 5: employment rises, and sales per worker improves

Panel A: firm-level employment and productivity.

- Median employment increases (from 1478.5 to 1849), but the median change is not strongly significant.
- Sales per employee increases: median change about +20.8 (thousand dollars per employee), statistically significant at conventional levels.
- Profit per employee (net profit or EBIT per employee) rises in median terms, but is statistically weaker than sales per employee.

Panel B: macro context. The paper complements the firm evidence with a time series on China's urban employment by ownership type, highlighting the gradual shift away from state/collective employment toward private and other ownership forms.

Table 5
Employment statistics

Panel A. Comparison of employment figures and employee productivity
The table presents the number of observations, the mean and median values of the employment level and employee productivity for the average of three years pre- and post-privatization periods, the mean and median change in the employment level and employee productivity value (post-pre), and the tests of significance of the median change. The Wilcoxon Z-test is employed to test for any significant change in the median value. The last column shows the number of positive versus negative changes and the proportion Z-test on if the proportion of positive change is greater than 50%. Employee productivity is in thousand dollars per employee.

Variable	Number of observation	Median (mean) before	Median (mean) after	Median (mean) change	Wilcoxon test	+ve/-ve Ratio (Prop. Z)
Employee	112	1478.50 (2427.69)	1849.00 (2857.18)	370.50 (429.49)	1.0649 (1.224)	63/49
Real sales/employee	112	105.86 (228.16)	126.07 (313.73)	20.81 (88.57)	1.8254* (2.721)**	70/42
RNP/employee	112	9.55 (19.76)	12.47 (25.19)	2.92 (15.43)	1.0259 (0.844)	60/52
REBIT/employee	97	14.871 (30.65)	17.618 (41.44)	2.747 (10.78)	0.4706 (1.653)*	56/41

Panel B. China urban year-end employment figures (in '0000 units)
The table presents the urban employment figures by production units. "SOE" are the state-owned units. "Collective" are the collective-owned units. "Coop" are the cooperative units. "Joint Own" are the joint ownership units. "Ltd. Corp" are the limited liability corporations. "Share Holding" are share holding corporations. "Private Ent" are private enterprises. "HK/M/T Funded" are units with funds from Hong Kong, Macau, and Taiwan. "Foreign Funded" are foreign funded units and "Self-employed" are self-employed individuals.

	SOE	Collective	Coop.	Joint own.	Ltd. corp.	Share holding	Private ent.	HK/M/T funded	Foreign funded	Self-employed
1978	7,451	2,048								15
1979	7,693	2,274								32
1980	8,019	2,425								81
1981	8,172	2,568								113
1982	8,630	2,651								147
1983	8,771	2,744								231
1984	8,837	3,216								339
1985	8,990	3,324								450
1986	9,333	3,421								483
1987	9,654	3,488								569
1988	9,983	3,527								659
1989	10,109	3,502								648
1990	10,346	3,549								614
1991	10,664	3,628								692
1992	10,889	3,621								740
1993	10,920	3,593								900
1994	10,980	3,511								1,225
1995	10,955	3,076								1,560

Table 6

Cross-sectional regression analysis on performance change
This table presents empirical results of the cross-sectional regression analysis on the full sample of privatized firms based on the following model:

$$\Delta PP_i = \alpha + \beta_1 \Delta GDPGR_i + \beta_2 SHSE_i + \beta_3 REG_i + \beta_4 REV_i + \beta_5 FRN_i + \beta_6 DUM96_i + \varepsilon_i$$

PP is the performance proxy. GDPGR is the real gross domestic product growth rate. The difference sign "Δ" is the average of the three-year post-privatization data minus the average of the three-year pre-privatization data of the variable in question, which captures the difference in mean growth of the variable before and after the privatization. SHSE is a dummy variable that takes the value of one if the firm is listed in the SHSE and zero if it is listed in SZSE. REG is a dummy variable that takes the value of one if a firm is in the regulated industry and zero otherwise. REV is a dummy variable that takes the value of one if the government retains more than 50% of shares in the firm and zero otherwise. FRN is a dummy variable that takes the value of one if a firm also issues foreign shares and zero otherwise. DUM96 is a dummy variable that takes the value of one if a firm was listed during 1996-1998 and zero otherwise. The *t*-statistics are in parentheses.

	ΔRNP	ΔREBIT	ΔROS	ΔEBITS	ΔRS	ΔTIE	ΔOCF/TO
Constant	0.4211 (1.82)*	-0.2808 (1.65)*	-0.0358 (-1.03)	-0.0444 (-1.78)*	0.2622 (1.74)*	-12.8672 (-6.33)	-0.1099 (-0.86)
ΔGDPGR	0.0326 (1.85)*	0.0736 (2.28)**	0.0034 (0.36)	-0.0005 (-0.06)	0.02509 (0.55)	0.3416 (0.03)	-0.0380 (-1.19)
SHSE	-0.1004 (-0.76)	0.0728 (0.74)	-0.0123 (-0.65)	-0.0032 (-0.13)	0.1415 (1.67)*	32.5747 (1.42)	-0.0521 (-0.51)
REG	0.5393 (2.61)**	0.3330 (1.93)*	0.0188 (0.44)	-0.0067 (-0.39)	0.2475 (1.32)	-8.0429 (-0.22)	-0.0618 (-0.75)
REV	-0.0731 (-0.28)	-0.1073 (-0.91)	0.0026 (0.53)	0.0291 (0.76)	-0.1517 (-1.67)*	-4.9633 (-0.10)	-0.124 (-1.01)
FRN	-0.0041 (-0.52)	-0.0081 (-0.67)	-0.0012 (0.43)	0.0006 (0.43)	-0.0051 (-1.43)	-27.1158 (-1.52)	-0.0053 (-0.12)
DUM96	0.3825 (2.28)**	0.4555 (4.14)**	0.0001 (0.02)	-0.0046 (-0.48)	0.4263 (4.36)**	16.5206 (0.82)	-0.0981 (-1.25)
Adj. R ²	0.0046	0.0286	-0.0068	-0.0044	0.0338	-0.0055	-0.0050
DW	2.0272	1.9325	1.9379	1.9486	1.9187	1.8729	1.9936
OBS	634	606	634	614	633	614	298

(*) (*) Denotes significance at the 10% (5%) level (two-tails).

Table 5 (continued)

	SOE	Collective	Coop.	Joint own.	Ltd. corp.	Share holding	Private ent.	HK/M/T funded	Foreign funded	Self-employed
1996	10,949	2,954								1,799
1997	10,766	2,817								1,919
1998	8,809	1,900								2,259
1999	8,336	1,652								2,414
2000	7,878	1,447								2,136

(*) (*) Denotes significance at the 10% (5%) level (two-tails).
*Source: China Statistical Yearbook (2001).

5 Controlling for macro trends: cross-sectional regression (Table 6)

The concern is that China's macro environment improved in the late 1990s, so the before-after changes may partly reflect *macroeconomic trends* rather than SIP.

5.1 Model: determinants of performance improvement (Eq. (2))

For each performance measure, the paper estimates:

$$\Delta PP_i = \alpha + \beta_1 \Delta GDPGR_i + \beta_2 SHSE_i + \beta_3 REG_i + \beta_4 REV_i + \beta_5 FRN_i + \beta_6 DUM96_i + \varepsilon_i, \quad (2)$$

where:

- ΔPP_i is the before-after change in one performance proxy (profit, output, leverage, etc.);

- $\Delta GDPGR_i$ captures the change in GDP growth between the pre and post periods;
- $SHSE_i$ is a Shanghai exchange indicator;
- REG_i indicates regulated industries;
- REV_i indicates revenue privatization (state share > 50%);
- FRN_i indicates foreign share issuance;
- $DUM96_i$ indicates listing after 1995 (1996–1998).

5.2 Table 6: interpretation of coefficients

Macro control matters, but it does not eliminate the SIP effect.

- $\Delta GDPGR_i$ is positively related to changes in real profit measures (e.g., significant for ΔRNP and $\Delta REBIT$), which is consistent with macro improvement boosting firms.
- Importantly, the intercept α remains positive and statistically meaningful for real profit changes and real sales changes, suggesting that the performance gains are not solely explained by macro trends.

Regulation and “control privatization” show up in the cross-section.

- REG_i is positive and significant in profit-change regressions, consistent with regulated industries receiving more support and/or enjoying market power.
- REV_i is associated with lower output growth (negative coefficient in ΔRS), consistent with the idea that majority state control can restrain expansion or distort incentives.
- $DUM96_i$ loads positively on profit and sales changes, suggesting later privatizations experienced larger improvements (possibly due to policy learning or a different reform stage).

6 Stock-market performance (Figure 1)

6.1 Portfolio construction

The paper forms equally weighted portfolios by listing year (IPO94, IPO95, ..., IPO98) and computes:

$$r_{p,t}^{raw} = \frac{1}{N_p} \sum_{i \in p} r_{i,t}, \quad (4)$$

$$r_{p,t}^{adj} = r_{p,t}^{raw} - r_t^{HS}, \quad (5)$$

where r_t^{HS} is the Hang Seng Index return (used as an external market benchmark).

6.2 Figure 1: interpretation

- Raw returns are generally positive after 1996, reflecting the overall bullish environment for Chinese-related equities.
- Adjusted returns still show large swings and strong cohort dependence, but there is *no clean evidence* that later cohorts systematically earn persistent abnormal returns.
- The pattern is consistent with an IPO/reform episode that changes ownership and scale, while stock returns remain heavily influenced by macro and market-wide sentiment.

7 Ownership, governance, and post-listing performance

7.1 Model: firm fixed-effects performance regression (Eq. (3))

To connect governance to outcomes, the paper estimates a pooled regression with firm fixed effects:

$$PP_{i,t} = \alpha_i + \beta_1 OWN_{i,t} + \beta_2 FRN_{i,t} + \beta_3 GDP_t + \beta_4 SIZE_{i,t} + \beta_5 LEVE_{i,t} + u_{i,t}, \quad (3)$$

where $PP_{i,t}$ is a performance measure (market-to-book, ROS, or EBIT/Sales), and $OWN_{i,t}$ is alternately the *state share* $ST_{i,t}$ or the *legal person share* $LP_{i,t}$ (estimated separately due to collinearity).

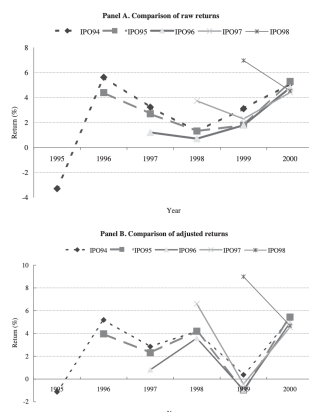


Fig. 1. Comparison of mean monthly returns. The figures show raw returns and market-adjusted returns of portfolios of stocks listed in different years.

Table 7
Pooled regression analysis of the impact of government ownership on firm performance after listing. This table presents empirical results of the pooled firm specific fixed-effects regressions:

$PP_{it} = \alpha_i + \beta_1 ST_{it} \text{ (or } LP_{it}) + \beta_2 FRN_{it} + \beta_3 GDP_t + \beta_4 SIZE_{it} + \beta_5 LEVE_{it} + u_{it}$
Performance proxies (PP) are MBR, ROS, and EBITs, respectively. MBR is the market value of equity divided by the book value of net assets. ROS is the net income per sales. EBITs is the operating income per sales. ST, LP, and FRN represent the fraction of shares of equity owned by the state, legal person, and foreigners respectively. GDP is the GDP growth rate. SIZE is the natural logarithm of total assets in RMB. LEVE is total liability divided by total assets. The pooled sample consists of 634 listed firms in both the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE) for 1994–2000. t -Values are in parentheses.

	MBR	ROS	EBITS
ST	-0.8518 (-1.73)*	-0.0535 (-1.17)	-0.0123 (-0.53)
LP	0.3631 (1.76)*	0.0786 (1.72)*	0.0362 (1.54)
FRN	1.6323 (1.93)*	-0.0324 (-0.63)	-0.0293 (-0.56)
GDP	0.7136 (8.37)**	0.7208 (8.62)**	0.0343 (7.86)**
SIZE	-2.4675 (-6.56)**	-2.4366 (-6.40)**	0.0800 (0.33)
LEVE	0.1567 (0.50)*	0.1561 (0.50)*	-0.0067 (-0.30)**
No. of obs.	2958	2958	2958
Adj. R ²	0.4477	0.4475	0.3744
DW	1.8760	1.8809	1.9908

Note: * and ** denote two-tailed significance at the 5% and 10% level, respectively.

f8.pdf

7.2 Table 7: what the fixed-effects results say

Market valuation (MBR).

- State ownership is negatively associated with valuation: ST enters with a negative coefficient (≈ -0.85) and marginal significance.
- Legal person ownership is positively associated with valuation: LP enters positively ($\approx +0.36$) and is also marginally significant.
- Foreign share fraction (FRN) is positively related to valuation (coefficients ≈ 1.6 – 1.7).

Accounting profitability (ROS and EBITs).

- *LP* is positively associated with ROS (significant at about the 10% level), while *ST* is not robustly related to ROS or EBITs.
- Leverage is strongly negatively related to profitability ratios, and GDP growth is positively related, as expected.

Interpretation. The valuation results fit the governance story: *state control* is perceived as value-reducing (political objectives, weaker monitoring), whereas *legal-person blockholding* can act as a monitoring/disciplining device.

8 Reverse causality and “who gets privatized” (Table 8)

A classic concern is selection: perhaps the government keeps more shares in inherently better firms, or foreign share issuance targets specific types of firms.

8.1 Model: determinants of state shares (Eq. (4))

The paper estimates:

$$ST_i = \alpha + \beta_1 SIZE_i + \beta_2 LEVE_i + \beta_3 ROS_i + \beta_4 SHSE_i + \sum_j \gamma_j IND_{i,j} + \varepsilon_i, \quad (4)$$

and similarly for $AVEST_i$ (average state share after listing).

8.2 Model: logit for foreign share issuance (Eq. (5))

Let FGN_i be an indicator for foreign share issuance. Then:

$$\Pr(FGN_i = 1 \mid X_i) = \frac{\exp\left(\alpha + \beta_1 SIZE_i + \beta_2 LEVE_i + \beta_3 ROS_i + \beta_4 SHSE_i + \sum_j \gamma_j IND_{i,j}\right)}{1 + \exp\left(\alpha + \beta_1 SIZE_i + \beta_2 LEVE_i + \beta_3 ROS_i + \beta_4 SHSE_i + \sum_j \gamma_j IND_{i,j}\right)}. \quad (5)$$

8.3 Table 8: interpretation

State retains more in larger and more leveraged firms.

- SIZE is strongly positively related to *ST*.
- LEVE is positively related to *ST*.
- Profitability (ROS) does *not* strongly predict *ST*.

This suggests that state retention is more about *strategic size and balance-sheet concerns* than ex-ante profitability.

Foreign share issuance is mainly about size, and negatively about leverage.

- Larger firms are more likely to issue foreign shares (strong positive SIZE coefficients in the logit).
- More leveraged firms are less likely to issue foreign shares (negative LEVE coefficients).

This is consistent with foreign investors selecting into larger, less financially constrained (less risky) issuers.

9 Short conclusion

9.1 What SIP achieved (in the data)

- **Scale expansion:** output (real sales) increases substantially post listing.
- **Earnings growth:** absolute profits (RNP, REBIT) increase in real terms.
- **But margins do not improve:** profitability ratios decline on average, especially outside regulated industries.
- **Debt burden rises:** debt-service ratios (TIE, OCF/TD) worsen, indicating leverage/interest pressure.
- **Labor productivity:** sales per employee rises; employment does not fall, consistent with partial reform.

Table 9

	1994	1995	1996	1997	1998	All years
<i>Panel A. Firms listed on the Shanghai Stock Exchange</i>						
Health Care	4	2	5	8	2	21
Media	1				1	2
Energy	1		5	2	4	12
Retail	14		13	5	1	33
Agriculture	2		8	9	7	26
Basic Materials	9	1	21	22	15	68
Utilities	3		4	5	2	14
Services	4	1	6	7	1	19
Property			1			1
Multi-industry	1		1		1	3
Manufacturing	22	6	24	15	12	79
Transport		1	5	4	2	12
Technology	2	1	7	7	2	19
Telecom			2	1	2	5
Total	63	12	102	85	52	314

Panel B. Firms listed on the Shenzhen Stock Exchange

Health Care	1		11	8	2	22
Media						0
Energy	1		5	3	2	11
Retail	2	1	9	8		20
Agriculture	4		5	12	6	27
Basic Materials	6	1	25	36	15	83
Utilities	5		1	3	1	10
Services	2	1	9	7	4	23

Table 9 (continued)

	1994	1995	1996	1997	1998	All years
Property	7	1	2	1		11
Multi-industry						0
Manufacturing	5	2	25	23	17	72
Transport	1	1		2	3	7
Technology	6	2	6	14	2	30
Telecom				4		4
Total	40	9	98	121	52	320

9.2 Governance lesson

Partial privatization is *not* the same as full privatization:

- Retained state ownership correlates with lower valuation.
- Legal-person ownership correlates with higher valuation (and somewhat higher profitability), consistent with monitoring incentives.

9.3 A limitation I keep in mind

The before–after design is informative but not a clean causal estimate: macro trends, selection into listing, concurrent reforms, and accounting changes still matter. The paper partially addresses this using GDP controls and ownership–selection regressions, but the identification is not quasi-experimental.

10 Appendix descriptive evidence in the paper (Tables 9–10)

10.1 Table 9 (Appendix): industry & listing-year composition

Table 9 is purely descriptive but useful for interpreting external validity:

- The sample is almost evenly split across exchanges (SHSE: 314 firms; SZSE: 320 firms).
- Listings cluster in 1996–1998 (SHSE: 87/105/100; SZSE: 85/99/83), consistent with an acceleration of SIP during the mid/late 1990s.
- Manufacturing dominates both exchanges (roughly 200 firms on each exchange), so “average effects” are heavily influenced by industrial firms.

10.2 Table 10 (Appendix): summary statistics for pooled regressions

Table 10 summarizes the pooled panel used in the ownership–performance regressions:

- **Market-to-book (MBR)** is very high on average (mean ≈ 5.24) and rises over time (mean ≈ 2.81 in 1994 to ≈ 7.44 in 2000), reflecting China’s high-valuation equity market in this era.
- **Ownership:** average state shares (ST) and legal-person shares (LP) are each around 30% on average, illustrating how “privatization” typically leaves large concentrated non-tradable blocks.
- **Foreign share fraction (FRN)** is small on average (mean $\approx 2.29\%$, median = 0), and it declines over time in the sample, consistent with foreign-share issuance being relatively limited and concentrated.
- **Scale and leverage:** sales increase over time; the liabilities-to-assets ratio (labeled “LA”) is roughly stable around 0.44.

In short, the pooled regressions in Table 7 are estimated in a setting with (i) very high valuations, (ii) large state/legal-person blocks, and (iii) small foreign stakes.

11 Conclusion

11.1 Summary

Sun and Tong (2003) study China’s *share-issue privatization* (SIP): SOEs are listed and sell a minority equity stake, while the state frequently retains large non-tradable blocks. The paper concludes that SIP is associated with **substantial real expansion and higher absolute earnings**,

but the **average profit margins and debt-service capacity deteriorate**, and the **ownership structure matters** for valuation and performance.

Empirically, comparing three-year pre- and post-listing averages:

1. **Absolute performance improves:** real net profit and real operating income rise markedly after listing. For the whole sample (Table 1),

Median RNP: $0.7277 \rightarrow 1.0925$ ($\Delta = +0.3648$), Median REBIT: $0.8149 \rightarrow 1.1386$ ($\Delta = +0.3237$).

These changes are highly statistically significant in signed-rank tests.

2. **Output expands strongly:** real sales increase sharply (Table 3),

Median real sales: $0.8827 \rightarrow 1.2489$ ($\Delta = +0.3662$).

The expansion is significantly larger when the state does *not* retain majority control (“control privatization”) relative to “revenue privatization” where the state keeps $> 50\%$.

3. **Profitability ratios decline on average:** despite higher absolute profits, margins fall (Table 2),

Median ROS: $0.1234 \rightarrow 0.1142$ ($\Delta = -0.0092$), Median EBIT/Sales: $0.1728 \rightarrow 0.1550$ ($\Delta = -0.0178$).

The margin decline is concentrated in non-regulated industries; regulated industries do not show the same deterioration.

4. **Debt-service capacity worsens (effective leverage rises):** (Table 4)

Median OCF/TD: $0.2360 \rightarrow 0.1800$ ($\Delta = -0.0560$), Median TIE: $6.2485 \rightarrow 4.7340$ ($\Delta = -1.5145$).

This is consistent with balance-sheet expansion and/or a higher interest burden post listing.

5. **Labor productivity rises:** sales per employee increases significantly (Table 5), while employment rises more modestly and not as robustly.
6. **Improvements are not purely macro-driven:** controlling for contemporaneous GDP growth and listing/industry dummies, intercepts remain positive in the earnings and sales-change regressions (Table 6), and later cohorts (1996–1998) exhibit significantly larger gains (DUM96 loads positively for Δ RNP, Δ REBIT, and Δ RS).

11.2 Ownership and governance conclusion (why partial privatization is not neutral)

A further conclusion is that **who owns the retained non-tradable blocks matters** (Table 7):

- **State ownership is valuation-reducing:** the state-share fraction (ST) is negatively asso-

ciated with market-to-book (MBR),

$$\hat{\beta}_{ST}^{MBR} \approx -0.8518 \quad (t \approx -1.73).$$

- **Legal-person ownership is valuation-enhancing and weakly profitability-enhancing:** the legal-person share (LP) loads positively on MBR and on ROS (both marginally significant).
- **Foreign shares raise valuation but do not reliably raise accounting performance:** FRN is positive for MBR but not robustly positive for ROS/EBITS, consistent with foreign investors being relatively passive (limited direct governance in this setting).

11.3 Economic intuition: why these patterns arise

The results are coherent under a simple “partial privatization with capital-raising” logic:

1. **Why profits and sales rise:** SIP is not just an ownership change; it is a *primary equity issuance* that injects capital into the firm. Listing also increases external scrutiny and can improve managerial incentives and access to financing. These forces naturally generate *scale expansion* (sales growth) and higher absolute earnings.
2. **Why margins fall:** rapid expansion, restructuring costs, and increased competition after listing can compress profit margins. In other words, firms can grow “bigger” faster than they grow “more efficient,” so ratios like ROS and EBIT/Sales fall.
3. **Why debt-service capacity worsens:** expansion often coincides with higher borrowing, and interest burdens may rise as firms lever up or as credit conditions tighten. Hence TIE and OCF/TD can decline even when profits rise in levels.
4. **Why ownership composition matters:** when the state remains dominant, political objectives and weaker monitoring can blunt efficiency gains and reduce valuation. In contrast, legal-person blockholders act more like monitoring institutions, improving governance at the margin.

11.4 Takeaway

China’s share-issue privatization is associated with strong post-listing growth in real sales and absolute profits, but average margins and debt-service capacity deteriorate, and governance frictions persist—state control is valuation-reducing while legal-person blockholding is valuation-enhancing.